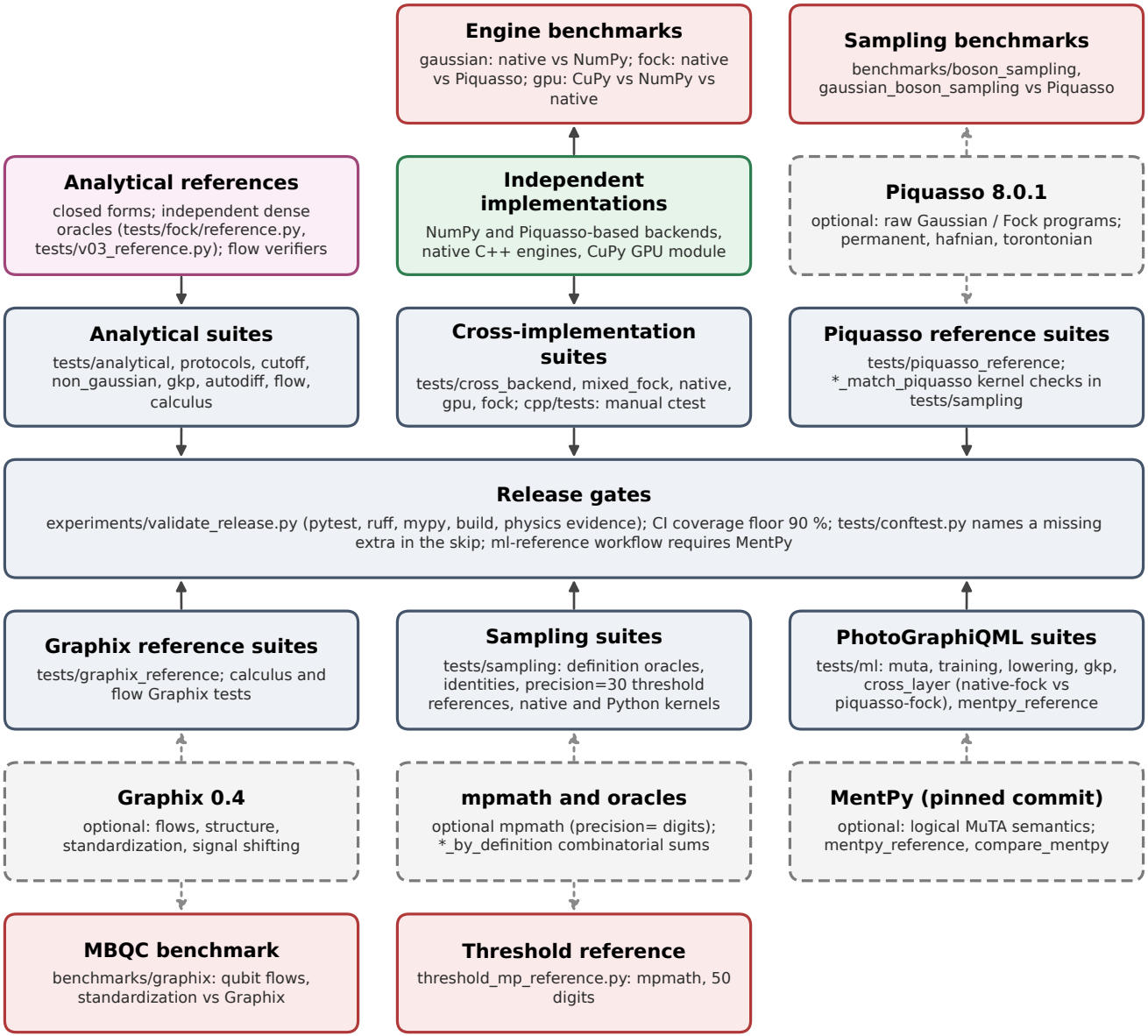H Software validation architecture